

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403868
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	McCabe; Kenneth et al.

Work vehicle with exterior debris removal system

Abstract

A work vehicle includes: a chassis; a cab carried by the chassis, the cab including a plurality of walls and a door movably coupled to at least one of the walls, the walls and the door together enclosing a cab interior, the door being movable between a closed position and an open position; a pressurized fluid source carried by the chassis and configured to output pressurized fluid; and at least one nozzle disposed outside the cab interior, directed at an area adjacent to the door, and fluidly couplable to the pressurized fluid source such that output pressurized fluid from the pressurized fluid source is directed towards the area adjacent to the door by the at least one nozzle to remove debris adjacent to the door.

Inventors:	McCabe; Kenneth (Romeoville, IL), Steuck; Sean (Greendale, WI)
Applicant:	CNH Industrial America LLC (New Holland, PA)
Family ID:	91667102
Assignee:	CNH Industrial America LLC (New Holland, PA)
Appl. No.:	18/091636
Filed:	December 30, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240217486 A1	Jul. 04, 2024

Publication Classification

Int. Cl.: B60S1/66 (20060101); F01N13/00 (20100101)

U.S. Cl.:

Field of Classification Search

CPC: B60S (1/66)

References Cited

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1086866	12/2000	EP	B60S 1/66

Primary Examiner: Ko; Jason Y

Background/Summary

FIELD OF THE INVENTION

(1) The present invention pertains to work vehicles and, more specifically, to work vehicles with cabs.

BACKGROUND OF THE INVENTION

(2) Work vehicles, such as tractors, are often used to perform a variety of tasks that involve debris. In agricultural applications, for example, dirt and residue from a field may be stirred up by the work vehicle and/or carried implements and collect on the work vehicle. Users typically control the work vehicle from a cab and debris inside the cab can make the cab an uncomfortable environment for the user. While many work vehicles have heating ventilation and air conditioning (HVAC) systems that include filters to reduce the amount of dirt, dust, and other debris that makes it into the cab, there are still instances where a significant amount of debris makes it into the cab and leads to user discomfort.

(3) What is needed in the art is a way to reduce the amount of debris that makes it into the cab of a work vehicle.

SUMMARY OF THE INVENTION

(4) Exemplary embodiments provided according to the present disclosure provide a work vehicle with one or more nozzles that are disposed outside of a cab of the work vehicle and directed towards an area adjacent to a door to remove debris adjacent to the door.

(5) In some embodiments provided according to the present disclosure, a work vehicle includes: a chassis; a cab carried by the chassis, the cab including a plurality of walls and a door movably coupled to at least one of the walls, the walls and the door together enclosing a cab interior, the door being movable between a closed position and an open position; a pressurized fluid source carried by the chassis and configured to output pressurized fluid; and at least one nozzle disposed outside the cab interior, directed at an area adjacent to the door, and fluidly couplable to the pressurized fluid source such that output pressurized fluid from the pressurized fluid source is directed towards the area adjacent to the door by the at least one nozzle to remove debris adjacent to the door.

(6) In some exemplary embodiments provided according to the present disclosure, a method of removing debris from an area adjacent to a door of a cab of a work vehicle is provided. The work vehicle includes a chassis, the cab being carried by the chassis and including a plurality of walls and the door coupled to at least one of the walls, the walls and the door together enclosing a cab

interior, the door being movable between a closed position and an open position. The work vehicle further includes a pressurized fluid source carried by the chassis and configured to output pressurized fluid. The method includes outputting pressurized fluid from the pressurized fluid source at the area adjacent to the door using at least one nozzle that is disposed outside the cab interior, directed at the area adjacent to the door, and fluidly coupled to the pressurized fluid source to remove the debris adjacent to the door.

(7) One possible advantage that may be realized by exemplary embodiments provided according to the present disclosure is that debris can be removed from the area adjacent to the door to reduce the risk that opening and closing of the door will draw debris into the cab interior.

(8) Another possible advantage that may be realized by exemplary embodiments provided according to the present disclosure is that debris can be removed from the area before the door will be opened and closed, to further reduce the risk of debris being drawn into the cab interior.

(9) Yet another possible advantage that may be realized by exemplary embodiments provided according to the present disclosure is that the pressurized fluid system may be a system of the work vehicle that performs a different function, such as an exhaust treatment system, to reduce or eliminate the need to include a further compressor or other element that would add significant weight to the work vehicle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For the purpose of illustration, there are shown in the drawings certain embodiments of the present invention. It should be understood, however, that the invention is not limited to the precise arrangements, dimensions, and instruments shown. Like numerals indicate like elements throughout the drawings. In the drawings:

(2) FIG. 1 illustrates a side view of an exemplary embodiment of a work vehicle included a cab provided according to the present disclosure in the form of a tractor;

(3) FIG. 2 illustrates a perspective view of the cab of FIG. 1 with a plurality of nozzles directed at an area adjacent to a door of the cab;

(4) FIG. 3 illustrates a fluid flow diagram of fluid from a pressurized fluid source to the nozzles of FIG. 2 with an actuatable valve that is in a first state;

(5) FIG. 4 illustrates the fluid flow diagram of FIG. 3 with the actuatable valve in a second state; and

(6) FIG. 5 illustrates a flow chart of an exemplary embodiment of a method of removing debris from the area adjacent to the door of the cab of FIGS. 1-2.

DETAILED DESCRIPTION OF THE INVENTION

(7) Referring now to the drawings, FIG. 1 illustrates a side view of one embodiment of a work vehicle **100**. As shown, the work vehicle **100** is configured as an agricultural tractor. However, in other embodiments, the work vehicle **100** may be configured as any other suitable work vehicle known in the art, such as various other agricultural vehicles, earth-moving vehicles, road vehicles, all-terrain vehicles, off-road vehicles, loaders, and/or the like.

(8) As shown in FIG. 1, the work vehicle **100** includes a pair of front wheels **102**, a pair of rear wheels **104**, and a chassis **106** coupled to and supported by the wheels **102**, **104**. A cab **108** is carried the chassis **106** and may house various control devices **110**, **112** (e.g., levers, pedals, control panels and/or the like) for permitting an operator to control the operation of the work vehicle **100**. Additionally, the work vehicle **100** may include an engine **114** and a transmission **116** mounted on the chassis **106**. The transmission **116** may be operably coupled to the engine **114** and may provide variably adjusted gear ratios for transferring engine power to the wheels **104** via a differential **118**.

(9) Moreover, the work vehicle **100** may also include an exhaust treatment system **200** for reducing

the amount emissions contained within the exhaust from the engine **114**. For instance, engine exhaust gas emitted by the engine **114** may be directed through the exhaust treatment system **200** to allow the levels of nitrous oxide (NOx) emissions contained within the exhaust to be reduced significantly. The cleaned or treated exhaust gases may then be expelled from the exhaust treatment system **200** into the surrounding environment via a clean exhaust **120**, which may include a pipe as illustrated in the work vehicle **100**. The exhaust treatment system **200** includes a catalytic reducer **201** that is coupled to an outlet **115** of the engine **114**, either directly or indirectly via one or more other elements of the exhaust treatment system **200**, and configured to produce cleaned exhaust gas from the engine exhaust gas.

(10) It should be appreciated that the configuration of the work vehicle **100** described above and shown in FIG. **1** is provided only to place the present subject matter in an exemplary field of use. Thus, it should be appreciated that the present subject matter may be readily adaptable to any manner of work vehicle configuration. For example, in an alternative embodiment, a separate frame or chassis may be provided to which the engine **114**, transmission **116**, and differential **118** are coupled, a configuration common in smaller tractors. Still, other configurations may use an articulated chassis to steer the work vehicle **100** or rely on tracks in lieu of the wheels **102**, **104**. Additionally, although not shown, the work vehicle **100** may also be configured to be operably coupled to any suitable type of work implement, such as a trailer, spray boom, manure tank, feed grinder, plow, and/or the like.

(11) Referring specifically now to FIG. **2**, the cab **108** is illustrated in further detail. The cab **108** includes a plurality of walls **210** and a door **220** that is movably coupled to at least one of the walls **210**. The walls **210** and the door **220** together enclose a cab interior, where the control devices **110**, **112** are housed. The door **220** is movable between a closed position, where the cab interior is generally enclosed, and an open position, where the cab interior is accessible by a user. The door **220** may, for example, be hinged to one of the walls **210** so the door **220** can swing outwardly relative to the cab interior and allow a user to leave the cab interior.

(12) In known work vehicles, debris tends to accumulate on various surfaces of the work vehicle. When the door opens and closes, debris can be drawn into the cab interior due to the movement of the door acting as a sail that aerosolizes debris and directs it into the cab interior. The amount of drawn-in debris can be significant, leading to user complaints due to the cab interior looking dirty as well as potentially irritating the user's respiratory system.

(13) To address the previously described issues with known work vehicles, and referring still to FIGS. **1-2** and now FIGS. **3-4** as well, the work vehicle **100** includes a pressurized fluid source that is carried by the chassis **106** and configured to output pressurized fluid and at least one nozzle, illustrated as a plurality of nozzles **230**, disposed outside the cab interior, directed at an area adjacent to the door **220**, and fluidly couplable to the pressurized fluid source such that output pressurized fluid from the pressurized fluid source is directed towards the area adjacent to the door **220** by the nozzle(s) **230** to remove debris adjacent to the door **220**. As used herein, "pressurized fluid" refers to any fluid, e.g., gas or liquid, that has a fluid pressure greater than atmospheric pressure so the pressurized fluid is expelled from the nozzle(s) **230** to remove debris adjacent to the door **220**. The area "adjacent to the door **220**" may be any area of the work vehicle **100** where debris may accumulate and then be drawn into the cab interior when the door **220** opens and closes. For example, the door **220** may include one or more seals along its perimeter that engage one or more walls **210** to seal the cab interior when the door **220** is closed. The nozzle(s) **230** may be directed towards the seal(s), as well as an area around the seal(s), so the pressurized fluid removes debris that collects on the seal(s) and the surrounding area to reduce the risk that debris is drawn into the cab interior. Thus, the nozzle(s) **230** directing pressurized fluid from the pressurized fluid source towards the area adjacent to the door **220** can significantly reduce the risk of debris entering the cab interior, especially when the door **220** is opened and closed.

(14) The pressurized fluid source can be provided in a variety of ways. In some embodiments, the

pressurized fluid source is a compressor or similar element that compresses fluid, such as air, and is fluidly coupled to the nozzle(s) **230** so the compressed/pressurized fluid can be output by the nozzle(s) **230** to remove debris adjacent to the door **220**. The compressor may, for example, be part of an HVAC system that controls the environment in the cab interior.

(15) In some embodiments, such as the embodiment illustrated in FIGS. **1-4**, the pressurized fluid source can be one or more elements of the exhaust treatment system **200**. Particularly, the cleaned exhaust gas that is output from the catalytic reducer **201** may be used as the output pressurized fluid that is output by the nozzle(s) **230** to remove debris adjacent to the door **220**. Specifically referring to FIGS. **3-4**, it is illustrated how the exhaust treatment system **200** may be fluidly couplable to the nozzle(s) **230** in order to remove debris. The catalytic reducer **201** may be coupled to an exhaust port **311** of the clean exhaust **120** via a first fluid path **310** of the clean exhaust **120**. The first fluid path **310** may be fluidly coupled to an actuatable valve **320**, which may be a proportional solenoid valve or other type of valve. In some embodiments, a cooler **240** is also provided that is configured to cool down cleaned exhaust gas from the catalytic reducer **201**, as is known.

(16) The actuatable valve **320** may be configured to change from a first state, illustrated in FIG. **3**, to a second state, illustrated in FIG. **4**. In the first state, the cleaned exhaust gas from the catalytic reducer **201** flows through the first fluid path **310** out the exhaust port **311** of the clean exhaust **120** so the cleaned exhaust gas vents to the atmosphere from the exhaust port **311**, similar to known vehicles. When the actuatable valve **320** changes from the first state to the second state, however, the cleaned exhaust gas tends to flow to a second fluid path **330** that is fluidly coupled to the nozzle(s) **230** so the cleaned exhaust gas is output by the nozzle(s) **230** and can remove debris adjacent to the door **220** as it is output to the atmosphere from the nozzle(s) **230**. In this respect, the clean exhaust **120** is fluidly coupled to the nozzle(s) **230** by the actuatable valve **320** and the second fluid path **330** being fluidly coupled to the first fluid path **310** of the clean exhaust **120**.

(17) In some embodiments, a compressor **340** is provided that is configured to further compress cleaned exhaust gas and increase the pressure of the cleaned exhaust gas for outputting by the nozzle(s) **230**. The compressor **340** may, for example, be fluidly coupled to the actuatable valve **320** on the side of the second fluid path **330** so only cleaned exhaust gas that will be output by the nozzle(s) **230** is further compressed by the compressor **340**. In some embodiments, the compressor **340** can also act similarly to the actuatable valve **320** and control when cleaned exhaust gas is output to the atmosphere through the exhaust port **311** or the nozzle(s) **230**.

(18) Generally, the actuatable valve **320** fluidly couples the pressurized fluid source (the exhaust treatment system **200**) to the nozzle(s) **230** through a fluid path (the second fluid path **330**) between the pressurized fluid source **200** and the nozzle(s) **230**, with the actuatable valve **320** defining a first state where the second fluid path **330** is closed to pressurized fluid from the pressurized fluid source and a second state where the second fluid path **330** is opened so the pressurized fluid is output by the nozzle(s) **230**. In this respect, the actuatable valve **320** can be controlled in order to control pressurized fluid being output by the nozzle(s) **230** and remove debris adjacent to the door **220**. To control the actuatable valve **320**, a controller **130** may be provided that is operably coupled to the actuatable valve **320**. The controller **130** is configured to output a first state signal so the actuatable valve **320** changes from the first state to the second state and output a second state signal so the actuatable valve **320** changes from the second state to the first state. As previously described, the actuatable valve **320** may be a proportional solenoid valve so the degree of openness of the actuatable valve **320** may be controlled, i.e., the actuatable valve **320** may be fully closed, partially open, or fully open. The actuatable valve **320** may be controlled by controlling current flow to the actuatable valve **320**. For example, in the first state the actuatable valve **320** may be de-energized so the second fluid path **330** is closed, i.e., fully or mostly closed. In the second state, on the other hand, the actuatable valve **320** may be energized so the second fluid path **330** is opened, partially or fully, and fluidly coupled to the pressurized fluid source **200**. In this respect, the controller **130**

outputting the respective signals can control the actuatable valve **320** to close and open the second fluid path **330** and control whether cleaned exhaust gas is output by the exhaust port **311** or the nozzle(s) **230**.

(19) While debris may collect adjacent to the door **220** at any time, it may mostly be an issue when the door **220** is opened and closed, which can draw the debris into the cab interior. Thus, in some embodiments, the controller **130** may be configured to control the actuatable valve **320** so pressurized fluid is output by the nozzle(s) **230** when the door **220** is likely to open and close. To effect this type of control, the controller **130** may be configured to receive a door open signal indicative that the door **220** will be moving from the closed position to the open position and to output the second state signal responsively to receiving the door open signal. The controller **130** can be configured to receive the door open signal from a variety of sources. For example, the door open signal may be output by a sensor **222** that is associated with a door handle **221** that allows the door **220** to open from the closed position. As a user utilizes the door handle **221** to open the door **220**, the controller **130** may receive the door open signal from the associated sensor **222** and output the second state signal so debris is removed. Such action can disperse some of the debris prior to the door **220** opening, reducing the amount of debris that is pulled into the cab interior.

(20) However, it may be possible to further reduce, if not eliminate, the amount of debris that is pulled into the cab interior during opening and closing of the door **220**. To further reduce the amount of debris pulled into the cab interior, the controller **130** may be configured to receive the door open signal from other sources that also indicate the door **220** is likely to be opened imminently. For example, the work vehicle **100** may include a service brake **140** that is configured to bring the work vehicle **100** to a stop upon activation and a service brake activator, which may be the control device **112**, inside the cab interior that is configured to output a service brake activation signal that activates the service brake **140**. The controller **130** may be operably coupled to the control device **112** so the controller **130** receives the service brake activation signal, which may also be the door open signal indicative that the door **220** will be moving from the closed position to the open position because the work vehicle **100** coming to a stop is indicative that the user may be planning to open the door **220** and exit the cab **108**. By tying the output of the pressurized fluid through the nozzle(s) **230** to activation of the service brake **140**, the debris can be removed from the area adjacent to the door **220** in advance of the user actually opening the door **220** so the debris is removed well before the door **220** opens, which can allow the debris to disperse from around the door **220**.

(21) In some embodiments, the controller **130** can additionally or alternatively be operably coupled to a speed sensor of the work vehicle **100** and configured to output the state signals as a function of the speed of the work vehicle **100**. For example, the controller **130** may be configured to output the first state signal to fully close the actuatable valve **320** when the work vehicle **100** travels at a defined travel speed and output the second state signal when the work vehicle **100** travels below the defined travel speed. In some embodiments, the controller **130** may be configured to output the second state signal at multiple defined speeds so more pressurized fluid is output through the nozzle(s) **230** as the speed of the work vehicle **100** decreases, which is indicative that a user may be opening and closing the door **220** soon. It should thus be appreciated that the controller **130** provided according to the present disclosure can be configured in a variety of ways to remove debris adjacent to the door **220** so the debris does not wind up in the cab interior during opening and closing of the door **220**.

(22) The nozzle(s) **230** may be provided in a variety of configurations and locations to remove debris adjacent to the door **220**. Many types of nozzles are known, and any suitable nozzle may be utilized according to the present disclosure. In the illustrated embodiment, there is provided a plurality of nozzles **230** that are disposed on a fender **150** of the work vehicle **100** that is carried by the chassis **106**. The fender **150** may be a curved form of fiberglass that is disposed adjacent to the cab **108** and the rear wheel **104** that is on the same side as the door **220**. The nozzles **230** may be

mounted to the fender **150** in any suitable manner. In some embodiments, the fender **150** includes one or more openings formed therethrough to receive fluid lines that fluidly couple the nozzle(s) **230** to the pressurized fluid source. A plenum or similar element may also be provided so only one opening in the fender **150** is needed to supply pressurized fluid to each of the nozzles **230**. Alternatively, one or more fluid lines to the nozzle(s) **230** may be provided without having to form openings through the fender **150**, e.g., by running the lines across an edge of the fender **150** or elsewhere. In some embodiments, the nozzle(s) **230** is directed at an area of the fender **150** that is between the nozzle(s) **230** and the door **220** to remove debris from the fender **150**, i.e., the area adjacent to the door **220** is a portion of the fender **150**.

(23) Further, the nozzles **230** may be directed in a variety of ways to remove debris adjacent to the door **220**. In some embodiments, the nozzles **230** are directed at the door **220** and the area adjacent to the door **220** so the output pressurized fluid tends to entrain the debris and then deflect off the door **220** (or bodywork adjacent to the door **220**) to remove the debris. However, it should be appreciated that the nozzles **230** can be directed at the area adjacent to the door **220** in other ways to remove debris. For example, the nozzles **230** may be disposed closely to the door **220**, i.e., in the area adjacent to the door **220**, and directed so the debris adjacent to the nozzles **230** is entrained by output pressurized fluid from the nozzles **230** and dispersed away from the door **220**. It should thus be appreciated that the nozzles **230** can be provided in a variety of ways and locations to remove debris adjacent to the door **220**.

(24) Referring now to FIG. 5, an exemplary embodiment of a method **500** of removing debris from the area of the work vehicle **100** adjacent to the door **220** is illustrated. The method **500** includes outputting **501** pressurized fluid from the pressurized fluid source **200** at the area adjacent to the door **220** using the at least one nozzle **230** to remove the debris in the area adjacent to the door **220**. When the actuatable valve **320** is provided, the method **500** may further include changing **502** the actuatable valve **320** to the first state from the second state and changing **503** the actuatable valve to the second state from the first state. The method **500** may further include determining **504** that the door **220** will be moving from the closed position to the open position and changing **503** the actuatable valve to the second state from the first state responsively to the determining **504**. The determining **504** may include, for example, activating the service brake **140**.

(25) From the foregoing, it should be appreciated that the work vehicle **100** provided according to the present disclosure utilizes one or more nozzles **230** to remove debris from an area adjacent to the door **220** of the cab **108**. By removing the debris with pressurized fluid, such as cleaned exhaust gas, the amount of debris that gets drawn into the cab interior during opening and closing of the door **220** can be reduced. It should thus be appreciated that exemplary embodiments provided according to the present disclosure can reduce the risk of the cab interior becoming dirty, which can increase user comfort and satisfaction.

(26) These and other advantages of the present invention will be apparent to those skilled in the art from the foregoing specification. Accordingly, it is to be recognized by those skilled in the art that changes or modifications may be made to the above-described embodiments without departing from the broad inventive concepts of the invention. It is to be understood that this invention is not limited to the particular embodiments described herein, but is intended to include all changes and modifications that are within the scope and spirit of the invention.

Claims

1. A work vehicle, comprising: a chassis; an engine carried by the chassis and comprising an outlet; a cab carried by the chassis, the cab comprising a plurality of walls and a door movably coupled to at least one of the walls, the walls and the door together enclosing a cab interior, the door being movable between a closed position and an open position; a pressurized fluid source carried by the chassis and configured to output pressurized fluid, wherein the pressurized fluid source comprises

an exhaust treatment system comprising a catalytic reducer that is coupled to the outlet and configured to produce cleaned exhaust gas from engine exhaust gas emitted by the engine; and at least one nozzle disposed outside the cab interior, directed at an area adjacent to the door, and fluidly couplable to the pressurized fluid source such that the pressurized fluid from the pressurized fluid source is directed towards the area adjacent to the door by the at least one nozzle to remove debris adjacent to the door, wherein the exhaust treatment system further comprises a clean exhaust that is fluidly coupled to the at least one nozzle and is configured to exhaust the cleaned exhaust gas to the atmosphere.

2. The work vehicle of claim 1, wherein the pressurized fluid is the cleaned exhaust gas.

3. The work vehicle of claim 1, further comprising an actuatable valve fluidly coupling the clean exhaust to the at least one nozzle, the actuatable valve defining a first state where the cleaned exhaust gas is exhausted to the atmosphere through an exhaust port of the clean exhaust and a second state where the cleaned exhaust gas is exhausted to the atmosphere through the at least one nozzle.

4. The work vehicle of claim 1, further comprising an actuatable valve fluidly coupling the pressurized fluid source to the at least one nozzle through a fluid path between the pressurized fluid source and the at least one nozzle, the actuatable valve defining a first state where the fluid path is closed and a second state where the fluid path is opened.

5. The work vehicle of claim 4, further comprising a controller operably coupled to the actuatable valve, the controller being configured to: output a first state signal so the actuatable valve changes to the first state from the second state; and output a second state signal so the actuatable valve changes to the second state from the first state.

6. The work vehicle of claim 5, wherein the controller is further configured to receive a door open signal indicative that the door will be moving from the closed position to the open position, wherein the controller is configured to output the second state signal responsively to receiving the door open signal.

7. The work vehicle of claim 6, further comprising a service brake configured to bring the work vehicle to a stop upon activation and a service brake activator inside the cab interior and configured to output a service brake activation signal that activates the service brake, wherein the service brake activation signal is also the door open signal.

8. The work vehicle of claim 1, further comprising a fender carried by the chassis, wherein the at least one nozzle is coupled to the fender.

9. The work vehicle of claim 1, wherein the at least one nozzle comprises a plurality of nozzles fluidly coupled to the pressurized fluid source.

10. A method of removing debris from an area adjacent to a door of a cab of a work vehicle, the work vehicle comprising a chassis, the cab carried by the chassis and comprising a plurality of walls and the door coupled to at least one of the walls, the walls and the door together enclosing a cab interior, the door being movable between a closed position and an open position, the work vehicle further comprising a pressurized fluid source carried by the chassis and configured to output pressurized fluid, the work vehicle further comprising an engine carried by the chassis and comprising an outlet, the pressurized fluid source comprising an exhaust treatment system comprising a catalytic reducer that is coupled to the outlet and configured to produce cleaned exhaust gas from engine exhaust gas emitted by the engine, the method comprising: outputting pressurized fluid from the pressurized fluid source at the area adjacent to the door using at least one nozzle that is disposed outside the cab interior, directed at the area adjacent to the door, and fluidly coupled to the pressurized fluid source to remove the debris in the area adjacent to the door, wherein the exhaust treatment system further comprises a clean exhaust that is fluidly coupled to the at least one nozzle and is configured to exhaust the cleaned exhaust gas to the atmosphere.

11. The method of claim 10, wherein the pressurized fluid is the cleaned exhaust gas.

12. The method of claim 10, wherein the work vehicle further comprises an actuatable valve fluidly

coupling the clean exhaust to the at least one nozzle, the actuatable valve defining a first state where the cleaned exhaust gas is exhausted to the atmosphere through an exhaust port of the clean exhaust and a second state where the cleaned exhaust gas is exhausted to the atmosphere through the at least one nozzle.

13. The method of claim 10, wherein the work vehicle further comprises an actuatable valve fluidly coupling the pressurized fluid source to the at least one nozzle through a fluid path between the pressurized fluid source and the at least one nozzle, the actuatable valve defining a first state where the fluid path is closed and a second state where the fluid path is opened.

14. The method of claim 13, further comprising: changing the actuatable valve to the first state from the second state; and changing the actuatable valve to the second state from the first state.

15. The method of claim 13, further comprising determining that the door will be moving from the closed position to the open position and changing the actuatable valve to the second state from the first state responsively to the determining.

16. The method of claim 15, wherein the work vehicle further comprises a service brake configured to bring the work vehicle to a stop upon activation, wherein the determining comprises activating the service brake.

17. The method of claim 10, wherein the work vehicle further comprises a fender carried by the chassis, wherein the at least one nozzle is coupled to the fender.

18. The method of claim 10, wherein the at least one nozzle comprises a plurality of nozzles fluidly coupled to the pressurized fluid source.

19. A work vehicle, comprising: a chassis; a cab carried by the chassis, the cab comprising a plurality of walls and a door movably coupled to at least one of the walls, the walls and the door together enclosing a cab interior, the door being movable between a closed position and an open position; a pressurized fluid source carried by the chassis and configured to output pressurized fluid; at least one nozzle disposed outside the cab interior, directed at an area adjacent to the door, and fluidly couplable to the pressurized fluid source such that the pressurized fluid from the pressurized fluid source is directed towards the area adjacent to the door by the at least one nozzle to remove debris adjacent to the door; an actuatable valve fluidly coupling the pressurized fluid source to the at least one nozzle through a fluid path between the pressurized fluid source and the at least one nozzle, the actuatable valve defining a first state where the fluid path is closed and a second state where the fluid path is opened; and a controller operably coupled to the actuatable valve, the controller being configured to: output a first state signal so the actuatable valve changes to the first state from the second state; and output a second state signal so the actuatable valve changes to the second state from the first state; wherein the controller is further configured to receive a door open signal indicative that the door will be moving from the closed position to the open position, and the controller is configured to output the second state signal responsively to receiving the door open signal.

20. The work vehicle of claim 19, further comprising a service brake configured to bring the work vehicle to a stop upon activation and a service brake activator inside the cab interior and configured to output a service brake activation signal that activates the service brake, wherein the service brake activation signal is also the door open signal.
